

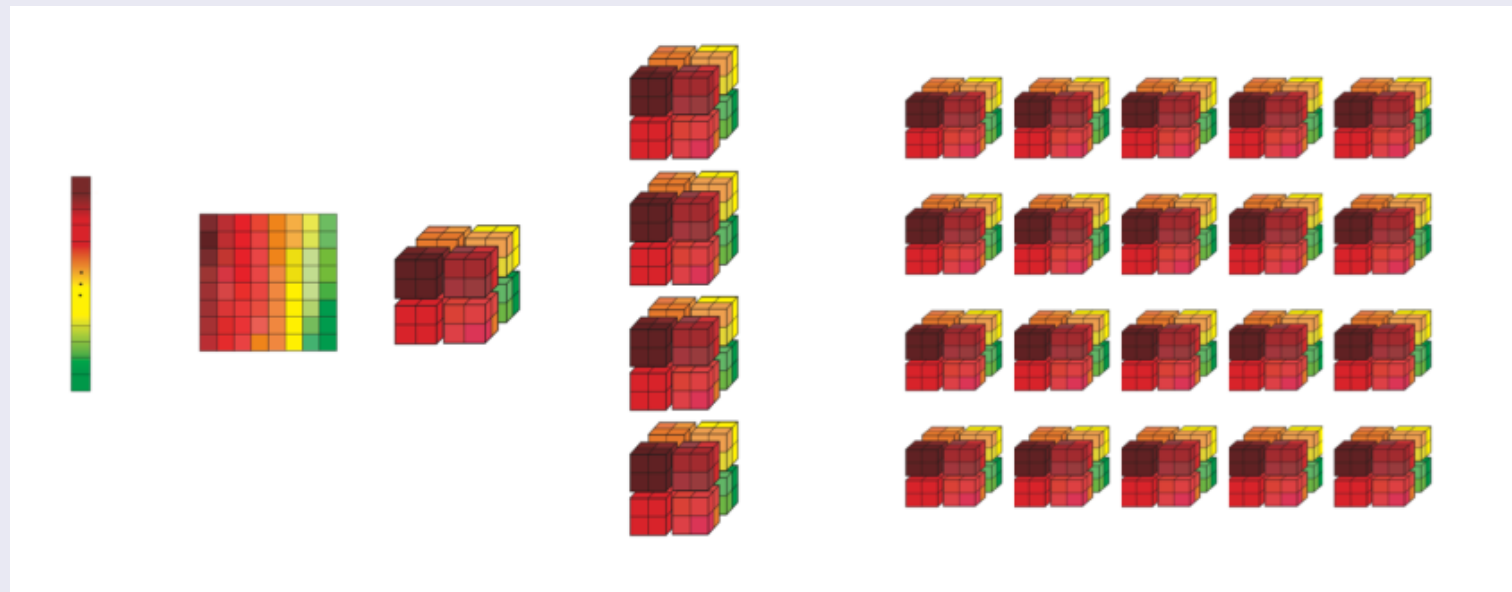
High-Dimensional Tensor Regression with Oracle Properties

Wenbin Wang, Yu Shi and Ziping Zhao

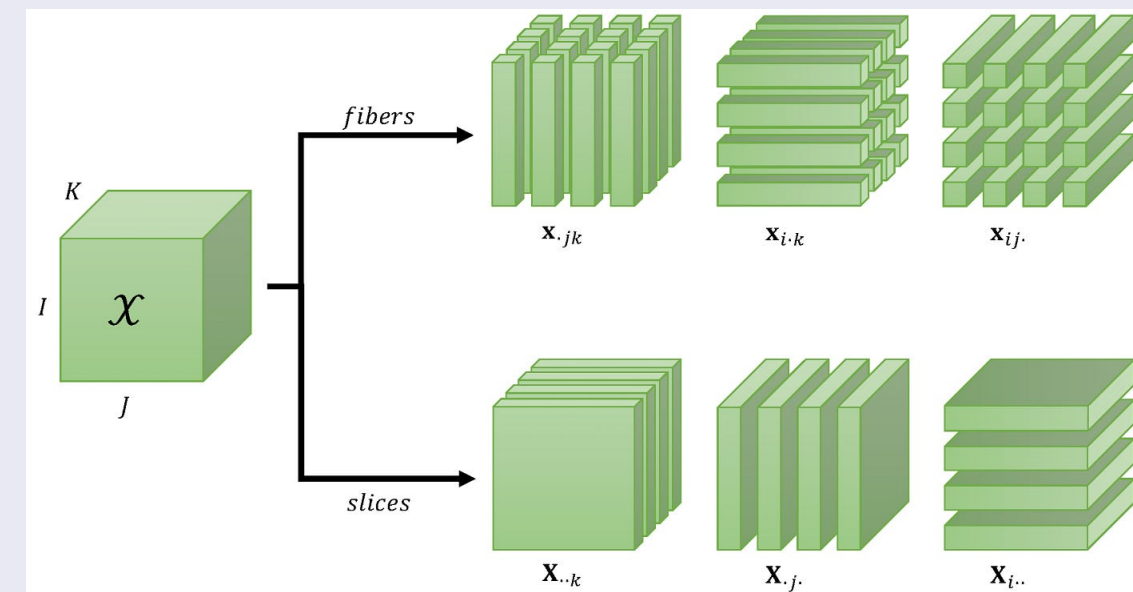
School of Information Science and Technology, ShanghaiTech University, Shanghai, China



Background



(a) Tensor are arrays with multiple directions.



(b) Tensor decomposition (CP Decomposition)

Tensor Regression:

$$\mathbf{y} = \langle \mathbf{X}, \mathbf{A}^* \rangle + \mathbf{E},$$

where $\mathbf{X} \in \mathbb{R}^{d_1 \times \dots \times d_M}$ is the predictor variable with $M \leq N$, $\mathbf{y} \in \mathbb{R}^{d_{M+1} \times \dots \times d_N}$ is the response variable, $\mathbf{A}^* \in \mathbb{R}^{d_1 \times \dots \times d_N}$ is the tensor coefficient, and $\mathbf{E} \in \mathbb{R}^{d_{M+1} \times \dots \times d_N}$ is the noise. $\langle \cdot, \cdot \rangle$ is the tensor contraction product between two tensors, where its $(i_{M+1}, \dots, i_N)$ -th entry is defined as

$$[\langle \mathbf{X}, \mathbf{A}^* \rangle]_{i_{M+1}, \dots, i_N} = \sum_{i_1=1}^{d_1} \dots \sum_{i_M=1}^{d_M} x_{i_1, \dots, i_M, i_{M+1}, \dots, i_N} a_{i_1, \dots, i_M, i_{M+1}, \dots, i_N}^*.$$

Proposed Problem

Given a collection of n samples $\left\{ \left(\mathbf{y}^{(i)}, \mathbf{x}^{(i)} \right) \right\}_{i=1}^n$, our goal is to estimate the unknown coefficient tensor $\mathbf{A}^*$ by solving the following regularized least squares estimation problem:

$$\min_{\mathbf{A} \in \mathbb{R}^{d_1 \times \dots \times d_N}} \left\{ \frac{1}{2n} \sum_{i=1}^n \left\| \mathbf{y}^{(i)} - \langle \mathbf{x}^{(i)}, \mathbf{A} \rangle \right\|_F^2 + \mathcal{R}_\lambda(\mathbf{A}) \right\},$$

where $\mathcal{R}_\lambda(\mathbf{A})$ is a structural regularization term. We consider a class of regularizers $\mathcal{R}_\lambda(\mathbf{A})$, which is defined based on a nonconvex penalty function $p_\lambda(\cdot)$ with parameter $\lambda \geq 0$. Assumptions on the non-convex penalty function $p_\lambda(\cdot)$ and $q_\lambda(t)$, where $q_\lambda(t)$ is the concave component of p_λ :

- There exists a constant $\nu > 0$ such that the penalty function satisfies $p'_\lambda(t) = 0$ for all $t \geq \nu$;
- $q_\lambda(t)$ is symmetric, i.e., $q_\lambda(-t) = q_\lambda(t)$ for all t ;
- $q'_\lambda(t)$ is monotone and Lipschitz continuous, i.e., for $t_2 \geq t_1$, there exists a nonnegative constant ζ such that $-\zeta \leq \frac{q'_\lambda(t_2) - q'_\lambda(t_1)}{t_2 - t_1} \leq 0$;
- $q_\lambda(t)$ and $q'_\lambda(t)$ pass through the origin, i.e. $q_\lambda(0) = q'_\lambda(0) = 0$;
- There exists a positive constant λ such that $|q'_\lambda(t)| \leq \lambda$ for all t .

Prototype examples of such regularizers include the SCAD and MCP, and we introduce them in the following.

$$\text{SCAD} : p_\lambda(t) = \begin{cases} \lambda|t|, & \text{for } |t| \leq \lambda, \\ -\frac{t^2 - 2a\lambda|t| + \lambda^2}{2(a-1)}, & \text{for } \lambda < |t| \leq a\lambda, \\ \frac{(a+1)\lambda^2}{2}, & \text{for } |t| > a\lambda, \end{cases}$$

where $a > 2$ is a tuning parameter.

$$p_\lambda(t) = \begin{cases} \lambda|t| - \frac{t^2}{2a}, & |t| \leq a\lambda, \\ \frac{a\lambda^2}{2}, & t > a\lambda, \end{cases}$$

for some constant $a > 1$.

Sparsity Regularization

A straightforward approach to induce sparsity within a tensor $\mathbf{A}$ is to enforce entry-wise sparsity. This strategy draws inspiration from the well-known Lasso regression. Specifically, the entry-wise sparsity regularizer is defined as

$$\mathcal{R}_\lambda(\mathbf{A}) = \sum_{i_1=1}^{d_1} \dots \sum_{i_N=1}^{d_N} p_\lambda(a_{i_1, \dots, i_N}).$$

In addition to promoting entry-wise sparsity, this regularization framework can be extended to incorporate other forms of sparsity in a general N -mode tensor, such as fiber-wise sparsity and slice-wise sparsity.

Low-rankness Regularization

In addition to promoting sparsity, encouraging low-rank structure in tensors has demonstrated significant benefits in various applications. We focus on mode-wise low-rankness, which involves penalizing the singular values of the mode- (k) unfoldings. Specifically, the mode-wise low-rankness regularizer is defined as

$$\mathcal{R}_\lambda(\mathbf{A}) = \sum_{i=1}^{\min\{d_k, \prod_{j \neq k} d_j\}} p_\lambda(\sigma_i(\mathbf{A}_{(k)})),$$

where $\sigma_i(\mathbf{A}_{(k)})$ denotes the i -th singular value of $\mathbf{A}_{(k)}$. Additionally, this framework can be extended to accommodate alternative forms of low-rankness regularization.

Main Results

Preliminaries: Define a local region as $\mathcal{C} = \{\mathbf{B} \in \mathbb{R}^{d_1 \times \dots \times d_N} \mid \mathcal{D}(\mathbf{B}, \mathbf{A}^*) \leq r\}$, where $\mathcal{D}(\cdot)$ denotes a distance function. For example, in the discussion of entry-wise sparsity,

$$\mathcal{D}(\mathbf{B}, \mathbf{A}^*) := \|\mathbf{B} - \mathbf{A}^*\|_F;$$

and in the discussion of mode-wise low-rankness, we define

$$\mathcal{D}(\mathbf{B}, \mathbf{A}^*) := \frac{\|\Pi_{\mathcal{F}_{\mathbf{A}^*}^\perp}(\mathbf{B})\|_{\text{nuc}}}{\|\Pi_{\mathcal{F}_{\mathbf{A}^*}}(\mathbf{B})\|_{\text{nuc}}},$$

where $\mathcal{F}_{\mathbf{A}^*}$ is a subspace associated with the unfolding of $\mathbf{A}^*$, $\mathcal{F}_{\mathbf{A}^*}^\perp$ is its orthogonal complement, and $\|\cdot\|_{\text{nuc}}$ denotes the nuclear norm.

Essential Assumptions:

- For any $\mathbf{A}, \mathbf{B} \in \mathcal{C}$, there exists a constant μ satisfying $\mu > 0$ such that $\mathcal{L}(\mathbf{B}) \geq \mathcal{L}(\mathbf{A}) + \langle \nabla \mathcal{L}(\mathbf{A}), \mathbf{B} - \mathbf{A} \rangle + \frac{\mu}{2} \|\mathbf{B} - \mathbf{A}\|_F^2$.
- For any $\mathbf{A}, \mathbf{B} \in \mathcal{C}$, there exists a constant L satisfying $L > 0$ such that $\mathcal{L}(\mathbf{B}) \leq \mathcal{L}(\mathbf{A}) + \langle \nabla \mathcal{L}(\mathbf{A}), \mathbf{B} - \mathbf{A} \rangle + \frac{L}{2} \|\mathbf{B} - \mathbf{A}\|_F^2$.
- Assume that the concatenation of vectorized covariates from n samples, denoted as $[\text{vec}(\mathbf{X}^{(1)})^\top, \dots, \text{vec}(\mathbf{X}^{(n)})^\top]^\top$, follows a multivariate Gaussian distribution with zero mean and covariance matrix Σ . We assume that there exist constants $\kappa \geq 1$ such that the eigenvalues of Σ satisfy $\kappa^{-1} \leq \sigma_{\min}(\Sigma) \leq \sigma_{\max}(\Sigma) \leq \kappa$.

Statistical Error

Sparsity: [Entry-wise sparsity] Suppose that Assumptions hold. If $\mu > \zeta$, $\lambda \asymp \sqrt{\frac{\log(d_1 d_2 \dots d_M)}{n}}$, and the true parameter tensor $\mathbf{A}^*$ satisfies

$$\min_{(i_1, \dots, i_N) \in \mathcal{S}_1} |a_{i_1, \dots, i_N}^*| \geq \nu,$$

then the optimal solution $\hat{\mathbf{A}}$ satisfies

$$\|\hat{\mathbf{A}} - \mathbf{A}^*\|_F \lesssim \sqrt{\frac{|\mathcal{S}_1|}{n}}.$$

Low-rankness: [Mode-wise low-rankness] Suppose that Assumptions hold.

If $\mu > \zeta$, $\lambda \gtrsim \frac{\sqrt{|\mathcal{S}_2|}}{n} \left\| \left[\mathbf{x}^* \left(\mathcal{E}^{(1:n)} \right) \right]_{(k)} \right\|_{\text{sp}}$, where $\|\cdot\|_{\text{sp}}$ is the spectral norm, and the true parameter tensor $\mathbf{A}^*$ satisfies

$$\min_i \sigma_i(\mathbf{A}_{(k)}^*) \geq \nu + \frac{2\sqrt{|\mathcal{S}_2|}}{n\mu} \left\| \left[\mathbf{x}^* \left(\mathcal{E}^{(1:n)} \right) \right]_{(k)} \right\|_{\text{sp}},$$

then the optimal solution $\hat{\mathbf{A}}$ satisfies

$$\|\hat{\mathbf{A}} - \mathbf{A}^*\|_F \lesssim \frac{\tau_k \sqrt{|\mathcal{S}_2|}}{n} \asymp \frac{\sqrt{|\mathcal{S}_2|}}{n},$$

where $\tau_k = \left\| \Pi_{\mathcal{F}} \left(\left[\mathbf{x}^* \left(\mathcal{E}^{(1:n)} \right) \right]_{(k)} \right) \right\|_{\text{sp}}$.

Numerical Simulations

We generate a set of independent covariate tensors $\{\mathbf{X}^{(i)}\}_{i=1}^n$, each entry independently drawn from a standard Gaussian distribution. The response variables are obtained according to tensor regression model, with additive independent Gaussian noise having zero mean and variance parameterized by η^2 . For all synthetic experiments, we employ 3rd-order tensors $\mathbf{A} \in \mathbb{R}^{d \times d \times d}$. For Figures 1 and 2, the noise parameter η is set to 0.1.

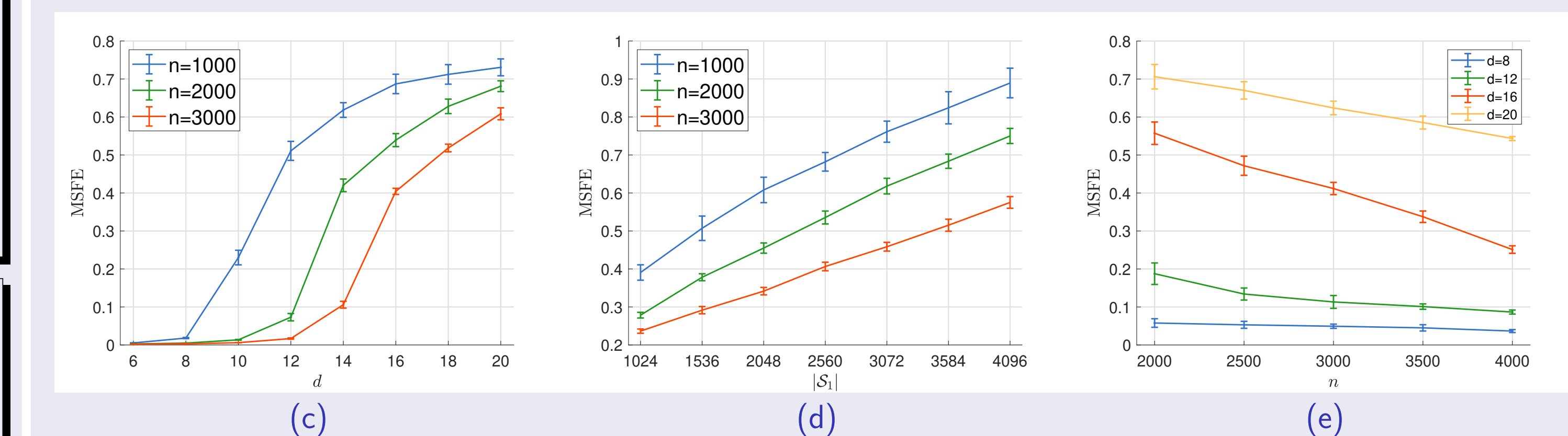


Figure: Entry-wise sparsity regularizer with the error bars of MSFE $\pm$ standard deviation.

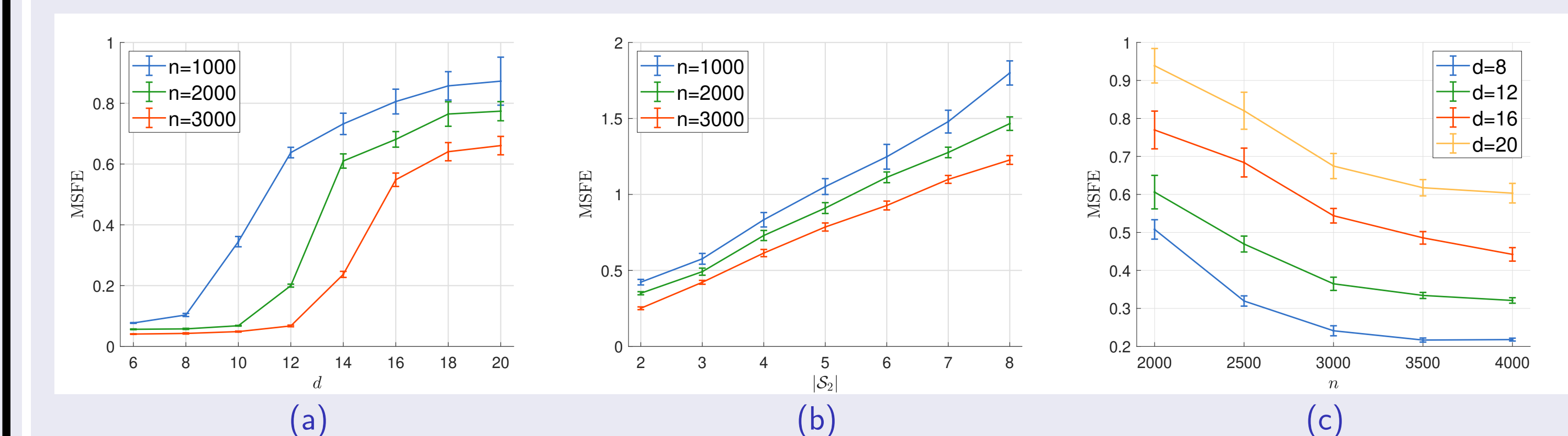


Figure: Mode-wise lowrankness regularizer with the error bars of MSFE $\pm$ standard deviation.